

Math 104 – Real Analysis

Comprehensive exercise solutions

Ross, *Elementary Analysis: The Theory of Calculus* (2nd ed.) ■ Fall 2026 ■ Keshav Ramamurthy

Adding a solution

To record a solved exercise, type under its chapter banner:

```
\exercise{8.6} \begin{solution} ... \end{solution}
```

Restate the problem first with `\begin{problem}[title]... \end{problem}`; structure proofs with `claim`, `idea`, `proofsketch`.

Contents

Chapter 1	Introduction	2
Chapter 2	Sequences	3
Chapter 3	Continuity	4
Chapter 4	Sequences and Series of Functions	5
Chapter 5	Differentiation	6
Chapter 6	Integration	7
Chapter 7	Capstone	8

Chapter 1 Introduction

Ross, §§1–6: natural, rational, and real numbers; the completeness axiom

Example 1.1 (1.1). We proceed via induction. Let $a_n = \sum_{i=1}^n i^2$.

Then $a_1 = 1 = \frac{1}{6}(1)(2)(3)$, so the base case holds.

Then, we proceed via induction. Suppose $a_i = \frac{1}{6}i(i+1)(2i+1)$ for $i \in \{1, 2, 3, 4, \dots, k\}$.

We will show that this implies that $a_{k+1} = \frac{1}{6}(k+1)(k+2)(2k+3)$

Chapter 2 Sequences

Ross, §§7–16: limits of sequences, limit theorems, monotone/Cauchy, subsequences, limsup/liminf, series

Chapter 3 Continuity

Ross, §§17–22: continuous functions, properties, uniform continuity, limits of functions

Chapter 4 Sequences and Series of Functions

Ross, §§23–27: power series, uniform convergence, term-by-term calculus

Chapter 5 Differentiation

Ross, §§28–31: derivative basics, Mean Value Theorem, L'Hospital, Taylor's Theorem

Chapter 6 Integration

Ross, §§32–36: Riemann integral, its properties, Fundamental Theorem of Calculus

Chapter 7 Capstone

Ross, §§37–38: exponents and logarithms; continuous nowhere-differentiable functions